

VAPO: Value-Model-Based RL for Advanced Reasoning 60.4 on AIME 2024

ByteDance Seed

Base Model: Qwen2.5-32B

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

- **Limitations of Value-Model-Free Methods (e.g., GRPO, DAPO):**
 - Compute advantage based on final trajectory reward and assign uniformly across all tokens.
- **Advantages of Value-Model-Based Methods:**
 - Credit assignment: Precise token-level credit assignment traces each action's impact on returns.
 - Variance reduction: Lower-variance estimates per token compared to Monte Carlo methods.
 - Generalization: Generalizes across samples

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

- **1. Value Model Bias**
 - Arises from long trajectories and bootstrapped learning.
- **2. Heterogeneous Sequence Lengths**
 - Different lengths require different bias-variance tradeoffs.
- **3. Reward Signal Sparsity**
 - Only the final <eos> token receives the reward signal.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Addressing Bias & Estimation:

- Value-Pretraining: Pre-trains value model before RL to reduce initial bias.
- Decoupled-GAE: Separates GAE computation to handle value estimation more robustly.

Addressing Heterogeneous Lengths:

- Length-adaptive GAE: Dynamically adjusts the λ parameter based on response length.

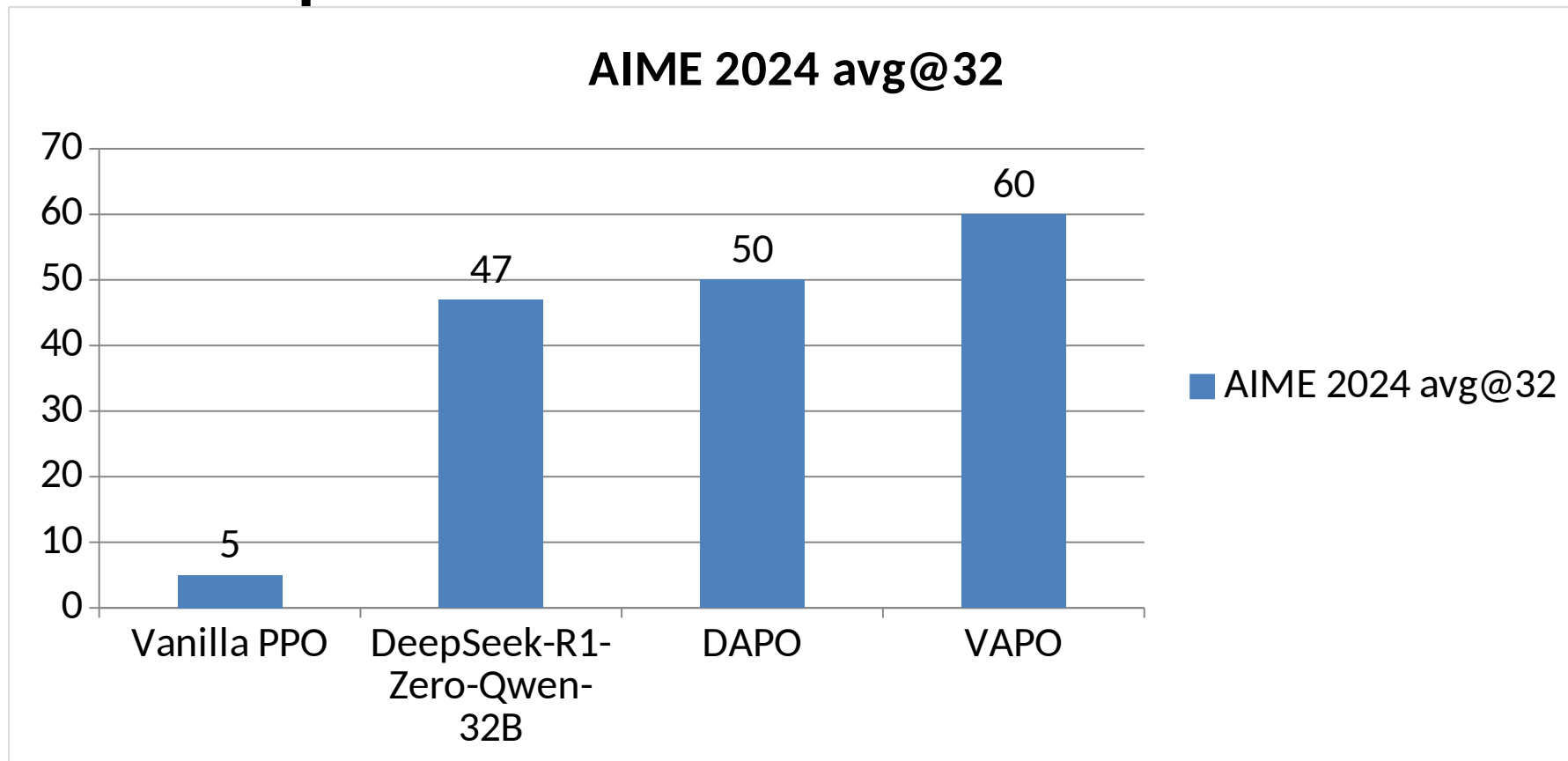
Addressing Exploration & Sparsity:

- Clip-Higher: Asymmetric clipping from DAPO encourages exploration.
- Token-level Loss: Normalizes loss at the token level rather than sequence level.
- Positive Example LM Loss: Self-imitation learning on correct trajectories.
- Group-Sampling: Samples multiple responses per prompt for better baseline estimation.

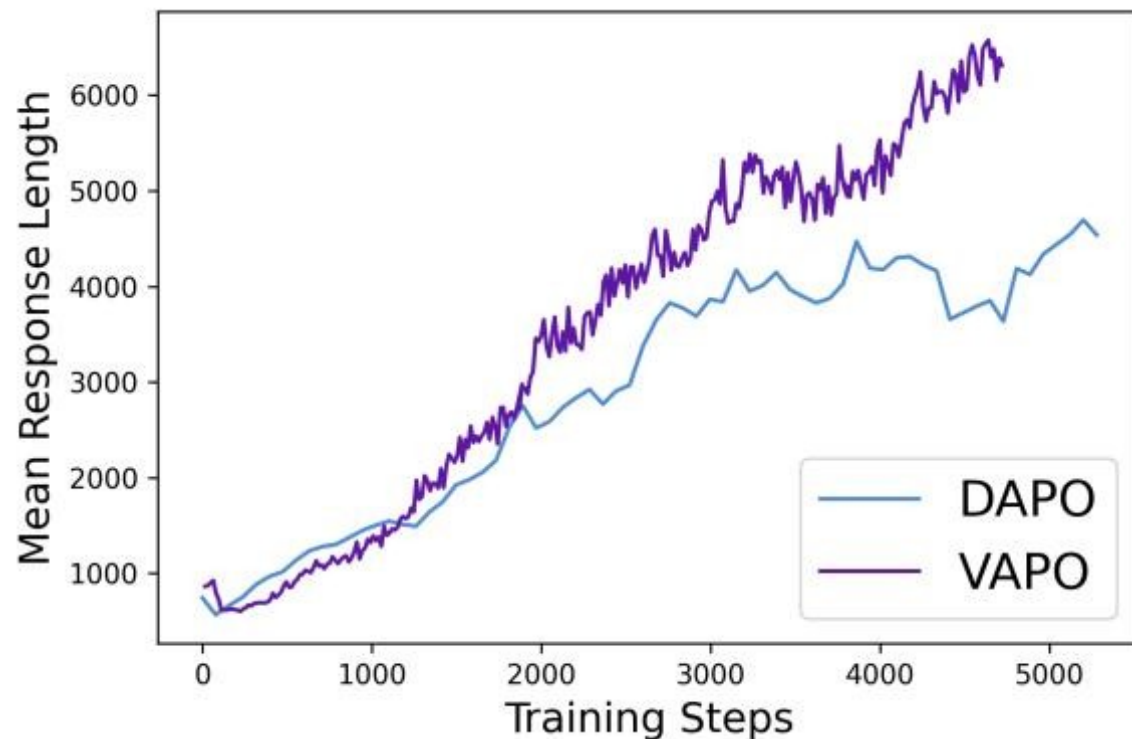
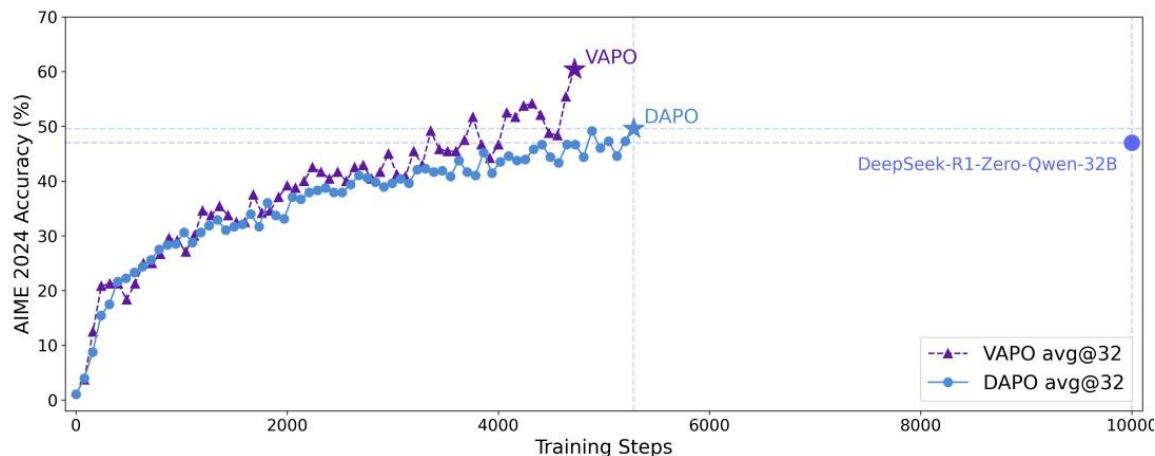
Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

- **Problem:** Heterogeneous sequence lengths unique to long-CoT reasoning.
- **Solution:** Dynamically adjust the λ parameter in GAE based on response length.
 - **Short Responses:**
 - Use lower λ (lower variance).
 - **Long Responses:**
 - Use higher λ (lower bias).
- **Result:** Optimizes the bias-variance tradeoff

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO



- VAPO reaches SOTA around 4,500 steps, while DAPO plateaus near 50% at 5,500 steps.
- VAPO generates longer reasoning chains (reaching ~6,000 tokens) with zero training crashes.

Ablation: Every Component Matters — Value-Pretraining Alone

Model	AIME24 avg@32
VAPO	60
VAPO w/o Group-Sampling	55
VAPO w/o Positive Example LM Loss	54
VAPO w/o Token-level Loss	53
VAPO w/o Clip-Higher	46
VAPO w/o Length-adaptive GAE	45
VAPO w/o Decoupled-GAE	33
VAPO w/o Value-Pretraining	11

Removing Value-Pretraining drops the score from 60 to 11. It is the single most critical component.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

- Value-model-based methods surpass value-free when challenges are addressed.
- Length-adaptive GAE is a critical innovation for heterogeneous lengths.
- Value-Pretraining is the single most important component (contributes 49 points).
- VAPO is stable with no training crashes across multiple runs, reaching SOTA in just 5,000 steps.